



Temporal EEG Signatures of Meditation Experience: Peak Brainwave Changes at 7 Minutes During Isha Yoga Breath Watching

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Abstract

Objectives Mind–body practices, such as meditation, have been shown to enhance mental well-being. Research shows that meditation influences brain function, with neurophysiological studies highlighting changes in alpha and theta frequency bands. However, when EEG changes begin after starting meditation and how they evolve over meditation state remain unknown.

Method To address this, we analyzed 128-channel EEG data during breath-watching meditation (10 min), a focused-attention practice, in meditation-naïve controls ($n=28$), novice meditators ($n=33$), and advanced meditators ($n=42$). Meditators belonged to the Isha Yoga tradition. We examined temporal dynamics in EEG power across different frequency bands (delta: 0.5–4 Hz; theta: 4–8 Hz; theta-alpha: 6–10 Hz; alpha: 8–12 Hz; beta1: 13–20 Hz; beta2: 20–30 Hz; and gamma1: 30–40 Hz) and groups. Threshold-Free Cluster Enhancement (TFCE)-based permutation statistics were used for statistical comparisons across multichannel EEG data.

Results All groups showed significant EEG changes—an increase in alpha, theta, and beta1 power and a decrease in delta and gamma1 power—emerging around 2–3 min after starting meditation and peaking between 7- and 10-min. Advanced meditators exhibited consistently higher theta and theta-alpha power at all time points.

Conclusions Our study suggests that the brain's response to meditation can be rapid and varies with practitioners' experience, potentially influencing cognitive and emotional processing in significant ways. Since meditation retreats may not be feasible for many individuals, brief meditation practices of seven min or more—delivered through digital platforms—could offer accessible, effective, and scalable solutions for improving mental well-being.

Preregistration This study is not preregistered.

Keywords Meditation · Brain · Breath-watching · EEG · Brain oscillations

Meditation is one of the most popular mind–body interventions, with its prevalence in the U.S. increasing from 7.5% in 2002 to 17.3% in 2022, and 200–500 million people practicing it globally (Davies et al., 2024; NCCIH, n.d.; United Nations, n.d.). It is commonly used to enhance attention, awareness, and emotion regulation (Dahl et al., 2015; Lutz et al., 2015; Tang et al., 2015). Importantly, meditation has been shown to reduce psychological stress, anxiety, and depression while enhancing joy, compassion, resilience, and well-being (Creswell, 2017; Galante et al., 2021; Goyal

et al., 2014; Sedlmeier et al., 2012; Upadhyay, Narayanan, et al., 2022a, b).

Meditation encompasses a wide variety of practices, broadly classified into focused-attention, open-monitoring, and loving-kindness meditation (Cahn & Polich, 2006; Fox et al., 2016; Lutz et al., 2008). Focused-attention practices include those practices that train attention on a particular object, such as the breath (Hölzel et al., 2011; Jha et al., 2007; Lutz et al., 2008). This practice of paying attention to the breath, called breath watching or mindful breathing, has been used in Buddhist and Yogic traditions to overcome suffering and gain insight into the true nature of the self (Nanamoli, 2010; Sadhasivam et al., 2021; Sadhguru, 2016). In addition to training attention, these practices also train awareness. Whenever the mind wanders, individuals need

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to cultivate awareness to notice mind-wandering and bring the attention back to the breath. Given that attention and awareness are often impaired in mental health disorders, their training is considered crucial for well-being (Dahl et al., 2015, 2020; Hölzel et al., 2011; Lutz et al., 2008, 2015; Tang et al., 2015).

The meditators in this study practiced breath-watching as taught in Isha Yoga, an international school of Yoga that provides tools for enhancing overall well-being (see methods for details on the tradition). Research on Isha Yoga practices has demonstrated beneficial outcomes, including reduced stress levels, heightened mindfulness, and enhanced well-being (Chang et al., 2022; Hariri et al., 2022; Malipeddi et al., 2024a, b; Upadhyay, Joshi, et al., 2022; Upadhyay, Narayanan, et al., 2022a, b). These practices have also shown benefits such as better cardiac sympathovagal balance (Muralikrishnan et al., 2012), heightened visual attention (Braboszcz et al., 2013), elevated brain-derived neurotrophic factor (BDNF) and cortisol awakening response (Cahn et al., 2017), increased anandamide levels (Sadhasivam et al., 2020), and more (Chandran et al., 2021; Malipeddi et al., 2024b). What are the neurodynamic features mediating the effects of Isha Yoga breath-watching practice? While some studies used fMRI (Vishnubhotla et al., 2021) and EEG (Braboszcz et al., 2017; Malipeddi et al., 2024b), the dynamic neural changes during the actual meditative state itself remain yet unclear. Addressing this gap in our knowledge is the goal of our study.

Electroencephalography (EEG), with its excellent temporal resolution, has been extensively used to study meditation (Cahn & Polich, 2006; Lee et al., 2018; Lomas et al., 2015). Research consistently shows an increase in power in alpha and theta frequency bands in the resting state (as a trait effect) and during meditation (as a state effect) (Cahn & Polich, 2006; Lomas et al., 2015). Changes in other bands, such as gamma, are reported only among advanced meditation practitioners (Braboszcz et al., 2017; Lutz et al., 2004; Sylapan et al., 2020). Further, meditators experience a state of “relaxed alertness” during meditation, where the mind is relaxed and alert at the same time. This state is reflected in the EEG by an increase in power in both low-frequency and high-frequency bands (Britton et al., 2014a; Dunn et al., 1999; Lomas et al., 2015; Malipeddi et al., 2024b).

Research so far has treated the brain during meditation as a static entity, while ignoring the potential dynamic modulations of brain activity during (the initiation of) the meditative state. It is common to analyze EEG data by examining short epochs, which are then averaged to obtain average activity (Braboszcz et al., 2017; Cahn & Polich, 2006; Lee et al., 2018; Lomas et al., 2015). These approaches effectively reduce the computational demands associated with analyzing extensive EEG data. However, they may inadvertently neglect the critical aspects of the time-to-onset of

neurophysiological effects and the intricate temporal dynamics of brain oscillation changes that occur throughout the meditation practice period. This oversight could limit our understanding of how these neural patterns evolve over time in response to meditation. Understanding these temporal dynamics could help optimize meditation duration for beginners.

Unlike previous approaches that focus on where brain activity changes during meditation, typically using spatially precise methods like fMRI (Fox et al., 2014, 2016; Ganesan et al., 2022), we aimed to explore when these changes occur, by tracking the temporal dynamics of neural activity throughout a meditation session. Specifically, we used EEG to monitor how power in different frequency bands (delta, theta, alpha, beta, and gamma) evolves over time during meditation. This dynamic, time-resolved approach allowed us to address three key questions as aims: (1) When do modulations in brain activity appear after starting meditation? (2) When do these modulations peak? and (3) Do these oscillatory changes differ between meditation-naïve controls, novice meditators, and advanced practitioners?

Method

Participants

Individuals with no prior meditation experience (meditation-naïve controls) were recruited ($n=28$; 16 females; age range: 24–51 years; mean age $\pm$ SD = 31.14 ± 6.38) from the local student community via WhatsApp advertisements, using a snowball sampling strategy. Meditators were recruited from the local Yoga Centres. Based on their level of meditation training, participants were categorized as either novice meditators ($n=33$; 14 females; age range: 23–54 years; mean age $\pm$ SD = 31.66 ± 7.64) or advanced meditators ($n=42$; 18 females; age range: 26–51 years; mean age $\pm$ SD = 35.57 ± 6.81). Advanced meditators were those who had completed Samyama, an intensive meditation retreat conducted by the Isha Foundation (Sadhasivam et al., 2021; Sadhguru, 2016). Novice meditators were those who had completed Shambhavi Mahamudra (Peterson et al., 2017; Sadhguru, 2016) but had not yet completed Samyama. The estimated lifetime hours of meditation practice of the groups are given in Table S1. Participants were matched for age, gender, and socio-economic status. Inclusion criteria for all participants included: healthy or under stable dosage of non-psychoactive medication (stable dosage period considered as a minimum of one month with no changes anticipated); able to read, write, and speak fluently in English, and able to comprehend and answer standard questionnaires. Exclusion criteria for all participants included: history of neurological disease, vision defects (uncorrected), auditory

deficits or serious physical disabilities; history of substance abuse/dependence; and history of major mental illness, any psychiatric medication, or psychotherapy. Novice and advanced meditators had to meet the following additional exclusion criteria: no current practice or history of practice of any other school of Yoga or meditation.

Meditation Tradition

Isha Yoga is a mind–body practice tradition with origins in India (Chandran et al., 2021; Malipeddi et al., 2024a, b; Sadhguru, 2016). It provides methods to enhance human well-being and consciousness. This tradition contains many practices, one of which is breath-watching (Malipeddi et al., 2024b), which is taught at both beginner (Chang, 2021; Peterson et al., 2017; Selvaraj et al., 2008) and advanced levels (Chandran et al., 2021; Sadhasivam et al., 2021). At a beginner level, it is part of the Shambhavi Mahamudra Kriya (offered as Inner Engineering) (Peterson et al., 2017; Selvaraj et al., 2008), a 21-min practice that includes pranayama (breathwork), chanting, and breath-watching, among others. One becomes eligible for Shoonya after learning Shambhavi. Shoonya is a practice of “conscious non-doing” (Braboszcz et al., 2017; Malipeddi et al., 2024b) that cultivates de-identification with body and mind, leading to non-dual states of awareness (Malipeddi et al., 2024b). At an advanced level, breath-watching is taught as part of the Samyama meditation retreat (Chandran et al., 2021), which requires learning Shambhavi, Shoonya meditation, and Hatha Yoga. Samyama is an 8-day advanced meditation retreat that needs participants to remain in total silence (Malipeddi et al., 2024b; Sadhasivam et al., 2021). Participation in the Samyama retreat requires a rigorous 60-day preparation involving specific vegan diet, regular practice of Shambhavi Mahamudra, Shoonya, and Hatha Yoga practices (Raman et al., 2023; Sadhasivam et al., 2021).

Procedure

The entire experimental paradigm consisted of multiple sessions conducted on the same day as one continuous session, including rest (4 min), pranayama (6 min), breath-watching meditation (15 min), Shoonya meditation (15 min), and a cognitive task (15 min) (Malipeddi et al., 2024b). In this study, we focused on investigating the ongoing temporal dynamics of oscillations during breath-watching meditation. Only the first 10 min of the breath-watching session were included for two reasons: (1) we had poor-quality recordings beyond 10 min (this study used a saline-based EEG system that reduces data quality beyond a certain time), and (2) many controls reported significant levels of drowsiness in the last few min of the practice. During this meditation session, participants

paid attention to the breath without identifying with their thoughts. Isha meditators performed breath-watching as they typically do in their regular practice, while meditation-naïve controls (who are first-time meditators) were taught this practice before the session. The following instructions were given to the control group: “*Pay attention to the natural movement of your breath. Whenever your mind wanders, simply notice it and gently bring your attention back to the breath.*” To promote engagement and adherence to the task, controls practiced breath-watching for a brief period prior to EEG recording and only proceeded once they confirmed understanding the task.

Measures

We performed additional analyses on our previously published data (Malipeddi et al., 2024b). EEG data were recorded using a 128-channel HydroCel Geodesic Sensor Net with the Geodesic EEG System 300 (Philips Neuro, USA) at a sampling rate of 1 kHz and 24-bit digitization. The procedure was conducted in a soundproof room with an ambient temperature of 25 °C and humidity controlled between 40 and 60%. Participants were instructed to wash their hair prior to the session to ensure high-quality EEG signals. Electrode impedance was kept below 50 k Ω , as recommended by the vendor.

Data Analyses

EEG Pre-processing

EEG data preprocessing was performed using the open-source MATLAB R2021b toolbox EEGLAB v2021.0 (Delorme & Makeig, 2004). The EEG data were resampled from 1000 to 250 Hz using EEGLAB's `resample` function and re-referenced to the average reference. The EEG data were first high-pass filtered above 0.5 Hz and low-pass filtered below 40 Hz using EEGLAB's ‘`pop_eegfiltnew`’ function with default filter order and a 25% transition bandwidth. A notch filter was also applied at 50 Hz to remove line noise. Bad channels and bad segments were automatically identified and rejected using the Artifact Subspace Reconstruction (ASR) plugin, with a threshold of five standard deviations. Subsequently, scalp muscle (EMG), ocular, and ECG artifacts were identified and removed using the ICALABEL plugin (90% confidence threshold) following independent component analysis (ICA) using the ‘extended infomax’ algorithm on the cleaned data. Finally, bad channels were interpolated using a spherical spline approach based on neighbouring electrodes.

EEG Analyses

Power spectral values were computed using MATLAB's bandpower function with a frequency resolution of 0.5 Hz and a Hamming window. Power was extracted for the following canonical frequency bands, defined as: delta (1–4 Hz), theta (4–8 Hz), theta-alpha (6–10 Hz), alpha (8–12 Hz), beta1 (13–20 Hz), beta2 (20–30 Hz), and gamma1 (30–40 Hz). These bands were selected based on prior EEG literature examining meditative and resting states (Cahn & Polich, 2006; Lomas et al., 2015). In addition to standard EEG bands, we included a theta-alpha (6–10 Hz) band based on prior findings suggesting that meditative states often involve coactivation of high-theta and low-alpha activity (Kakumanu et al., 2018). Including this band allowed us to more sensitively capture meditation-related oscillatory dynamics that may not be reflected in traditional frequency band boundaries. We did not include gamma2 (40–110 Hz) band as this is usually contaminated with major muscle artifacts (Muthukumaraswamy, 2013).

We used the first 30 s of breath-watching (10 min) as the baseline to explore the time-to-onset of neurophysiological effects and the temporal dynamics of the practice across all three groups. Time-to-onset was defined as the time point at which statistically significant brain changes occur. For within-group comparisons, differences in absolute power between the first 30 s of data and successive 1-min segments (non-overlapping non-sliding windows) were calculated. Each 1-min time point represented cumulative power until that point. For between-group comparisons, differences in absolute power among the three groups were calculated at 0.5, 3, 6, and 9 min. While line plots in Fig. 4 capture the temporal evolution of power changes at one-minute resolution, plotting topographical maps at the same resolution resulted in visual clutter. To enhance clarity and avoid repetition, we chose to present cumulative power at three-minute intervals in the topographical maps.

Statistical Analyses

Threshold-Free Cluster Enhancement (TFCE)-based permutation statistics were used for statistical comparisons across multichannel EEG data (Pernet et al., 2015). Surrogate data generated during the permutation process, along with non-parametric bootstrapping, was employed to estimate statistical significance and correct for multiple comparisons. TFCE enhances the *t*-values at each electrode by integrating cluster mass over all possible thresholds, resulting in TFCE scores. *P*-values were computed through permutation testing of the maximum TFCE scores, thereby controlling for the family-wise error rate (FWER). Robust Student's *t*-values were calculated for each data point within participants prior to applying cluster permutation tests. Channel clusters were

identified following standardized guidelines (as detailed here in Rahul, 2023). When using traditional clustering methods to correct for multiple comparisons, a cluster-forming threshold must be defined. However, different thresholds can yield clusters with varying spatial or temporal extents, potentially influencing the sensitivity of the subsequent permutation test. To address this limitation, the Threshold-Free Cluster Enhancement (TFCE) method was introduced by Smith and Nichols (Smith & Nichols, 2009). EEG effects often span neighbouring sensors and time points. TFCE enhances signal-like clusters while suppressing noise by considering both the extent (cluster size) and height (effect magnitude) of the signal. As a result, it favours broad and strong activations, which are more likely to reflect genuine EEG effects. Statistical comparisons were implemented using the LIMO toolbox (Pernet et al., 2011), employing robust *t*-statistics with 2000 permutations, with corrections for multiple comparisons across electrodes using cluster statistics. Level of significance was set at $\alpha < 0.05$. Finally, the percent bent robust correlation (pbcor) was employed (Mair & Wilcox, 2020) to assess the relationship between averaged band powers (e.g., Theta and Gamma1 PSD), providing resistance to outliers, while boxplots were generated to visualize the distribution of the mean power spectral density (PSD) across electrodes for each frequency band, group, and time point.

Results

Within-group Comparisons of Dynamic Changes Over Time during Meditation Practice

Neurophysiological effects emerged around 2–3 min in meditation-naïve controls, novices, and advanced meditators (Table 1). Compared to the first 30 s of breath-watching, all three groups showed significantly increased theta (4–8 Hz) ($p < 0.05$), alpha (8–12 Hz) ($p < 0.05$), theta-alpha (6–10 Hz) ($p < 0.05$) and beta1 (13–20 Hz) ($p < 0.05$) power and decreased delta (1–4 Hz) ($p < 0.05$) and gamma1 (30–40 Hz) ($p < 0.05$) power at 2–3 min. No changes were observed in the beta2 (20–30 Hz) band in any group. We observed different times to onset of effects across frequency bands and groups. For all three groups, delta and gamma1 power decreased with time, whereas theta, theta-alpha, alpha, and beta1 power increased with time. The maximal effects in power were observed in the 7–10-min duration in the different groups. These results are summarized in Table 1.

For the control group (Fig. 1), significant time-to-onset of effects was observed at 2 min in the delta (central, left parieto-occipital, and right frontal regions) ($p < 0.05$), theta (left parieto-occipital and right frontal regions) ($p < 0.05$), theta-alpha (whole brain) ($p < 0.05$), alpha (whole brain except

Table 1 Temporal and power dynamics of EEG effects across frequency bands and groups during breath-watching meditation

Frequency band	Group	Time-to-Onset (min)	Maximum effects (min)	Power change from first 30 s (Increase/Decrease) ($p < 0.05$)
Delta (0.5 – 4 Hz)	Controls	2	10	Decrease
	Novice meditators	1	10	Decrease
	Advanced meditators	2	9	Decrease
Theta (4 – 8 Hz)	Controls	2	8	Increase
	Novice meditators	2	7	Increase
	Advanced meditators	1	10	Decrease (at 1 min) and then increase
Theta-alpha (6 – 10 Hz)	Controls	2	10	Increase
	Novice meditators	2	10	Increase
	Advanced meditators	2	10	Increase
Alpha (8 – 12 Hz)	Controls	2	10	Increase
	Novice meditators	2	10	Increase
	Advanced meditators	2	9	Increase
Beta1 (13 – 20 Hz)	Controls	2	7	Increase
	Novice meditators	1	9	Increase
	Advanced meditators	2	8	Increase
Beta2 (20 – 30 Hz)	Controls	-	-	No change
	Novice meditators	-	-	No change
	Advanced meditators	-	-	No change
Gamma1 (30 – 40 Hz)	Controls	2	10	Decrease
	Novice meditators	3	10	Decrease
	Advanced meditators	3	10	Decrease

right central) ($p < 0.05$), beta1 (whole brain) ($p < 0.05$) and gamma1 bands (whole brain) ($p < 0.05$). For reference on anatomical regions and electrode locations, see here (Krol, 2020).

In the novice meditator group (Fig. 2), time-to-onset patterns differed notably from controls, both in timing and topography. Effects emerged earlier in several bands, with significant changes observed at 1 min in the delta band (whole brain except right posterior regions) ($p < 0.05$), and at 2 min in the theta ($p < 0.05$), theta-alpha ($p < 0.05$), and alpha bands ($p < 0.05$) (whole brain). The beta1 band showed effects at 1 min (fronto-central and central regions) ($p < 0.05$), and gamma1 effects appeared at 3 min (whole brain) ($p < 0.05$).

Finally, in the advanced meditator group (Fig. 3), significant time-to-onset of effects were observed at 2 min in the delta band (whole brain) ($p < 0.05$), 1 min in the theta band (whole brain) ($p < 0.05$), 2 min in the theta-alpha band (posterior regions) ($p < 0.05$), 2 min in the alpha band (whole brain) ($p < 0.05$), 2 min in the beta1 band (posterior regions) ($p < 0.05$), and 3 min in the gamma1 band (whole brain) ($p < 0.05$). A unique temporal pattern was observed in the theta band: power initially decreased at 1 min, followed by an increase from 2 min onward — a pattern not seen in the other groups.

The time-to-onset of EEG effects following breath-watching meditation differed notably between groups. While controls exhibited changes across all frequency bands beginning at 2 min, meditators showed more varied patterns, with some effects emerging earlier — such as delta in novice meditators and theta in advanced meditators at 1 min — and others emerging later, such as gamma1 at 3 min in both meditator groups. In addition to these temporal differences, the spatial distribution of effects also varied. This was particularly evident in the theta band: controls showed increased power in a few electrodes at 2 min; novice meditators exhibited a widespread increase across all electrodes at the same time point; and advanced meditators displayed a unique trajectory, with a decrease in power at 1 min followed by a global increase at 2 min. These findings suggest that the brain's response to meditation is not uniform but varies systematically with the level of meditative expertise, reflecting differences in how quickly and broadly the brain engages with the meditative state.

In summary, all groups showed dynamic changes in the power of different frequency including delta (decrease), theta and alpha (increase) and gamma (decrease) which started around 2–3 min and peaked around 7 min. The exact timing, power bands, and electrodes differed between the three groups, though, showing either earlier or later changes.

Temporal dynamics during breath-watching meditation in first-time meditators

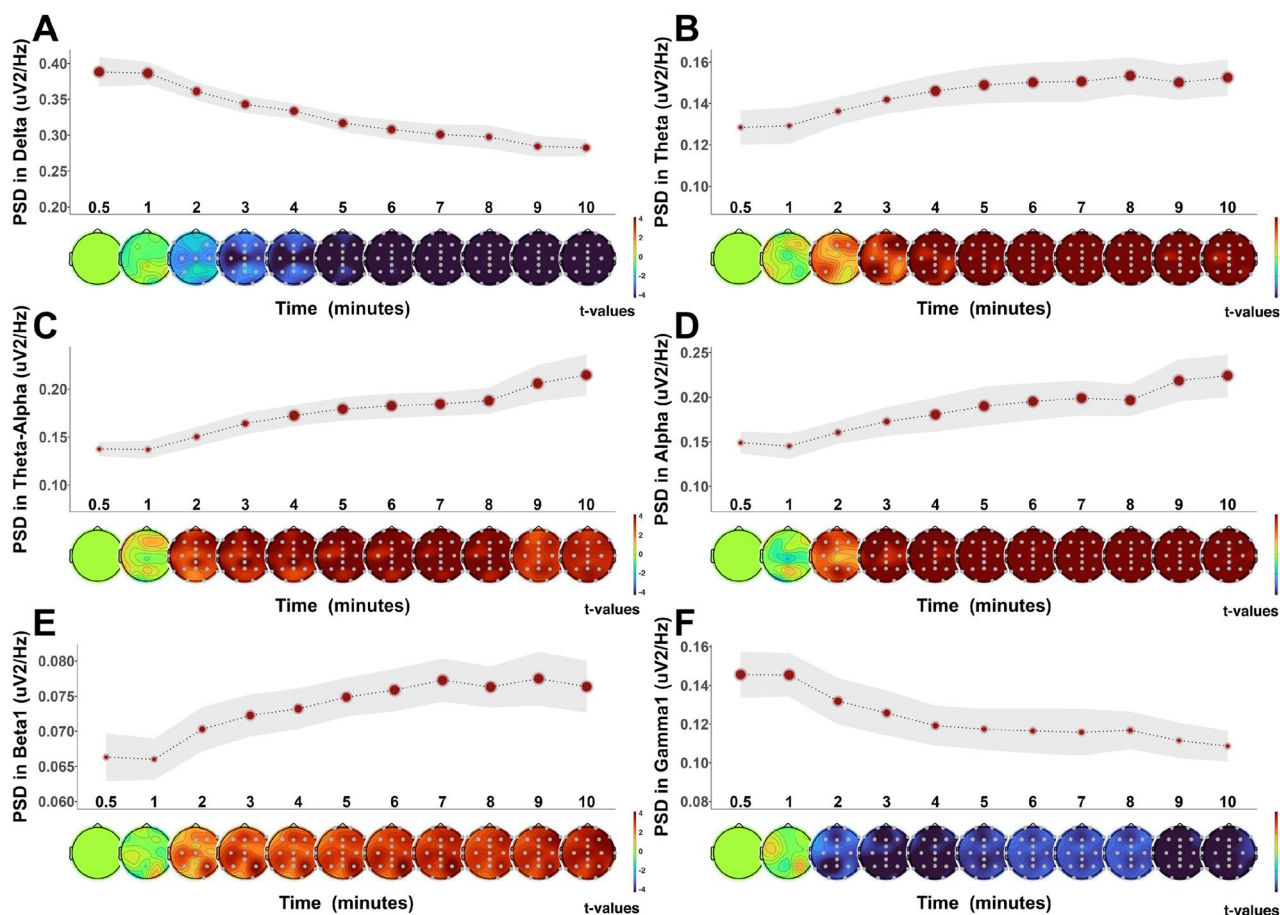


Fig. 1 Temporal dynamics in meditation-naïve controls: Time-to-onset and temporal dynamics during breath-watching meditation in first-time meditators (meditation-naïve controls) (CNT) are illustrated. Power-spectral density (PSD) values in different frequency bands (A: Delta; B: Theta; C: Theta-alpha; D: Alpha; E: Beta1; and F: Gamma1) show time-to-onset of effects around a 2–3-min dura-

tion. The size of the dots corresponds to the magnitude of the values. The grey bar indicates standard error. The lower head plots are the t-value topographic plots (compared to the first 30 s) that highlight electrodes with statistically significant differences (dots, $p < 0.05$) in power. *P*-values have been corrected for multiple comparisons using cluster-based statistics

We therefore conducted between-group comparisons (given below).

Within-Group Comparisons with Eyes-closed Rest as the Baseline

We also conducted the same analysis, as above, using the eyes-closed resting state as the baseline (Figs. S1–S7). In the delta band (Fig. S1), overall power during breath-watching was higher than rest across all three groups; however, the power values for both states converged over time. In the theta band (Fig. S2), advanced and novice meditators showed lower power values during breath-watching than rest for the first two min, followed by a gradual increase that peaked around 8 min. Controls began breath-watching with power values similar to rest and then

showed a gradual increase, with effects peaking around 7 min. In the theta-alpha band (Fig. S3), all three groups initially showed lower power during breath-watching than rest; however, as the session progressed, the power values converged. A similar pattern was observed in the alpha (Fig. S4) and beta1 bands (Fig. S5). There were no differences in power between rest and the beta2 band at any time point (Fig. S6). In the gamma1 band (Fig. S7), both advanced and novice meditators showed higher power during breath-watching than the rest in the early stages of practice. As meditation progressed, power returned to resting state levels at approximately 5 min in advanced meditators and 2 min in novice meditators. Controls showed no differences between breath-watching and rest at any time point. The above results show that the choice of the baseline influences findings and how they are interpreted.

Temporal dynamics during breath-watching meditation in novice meditators

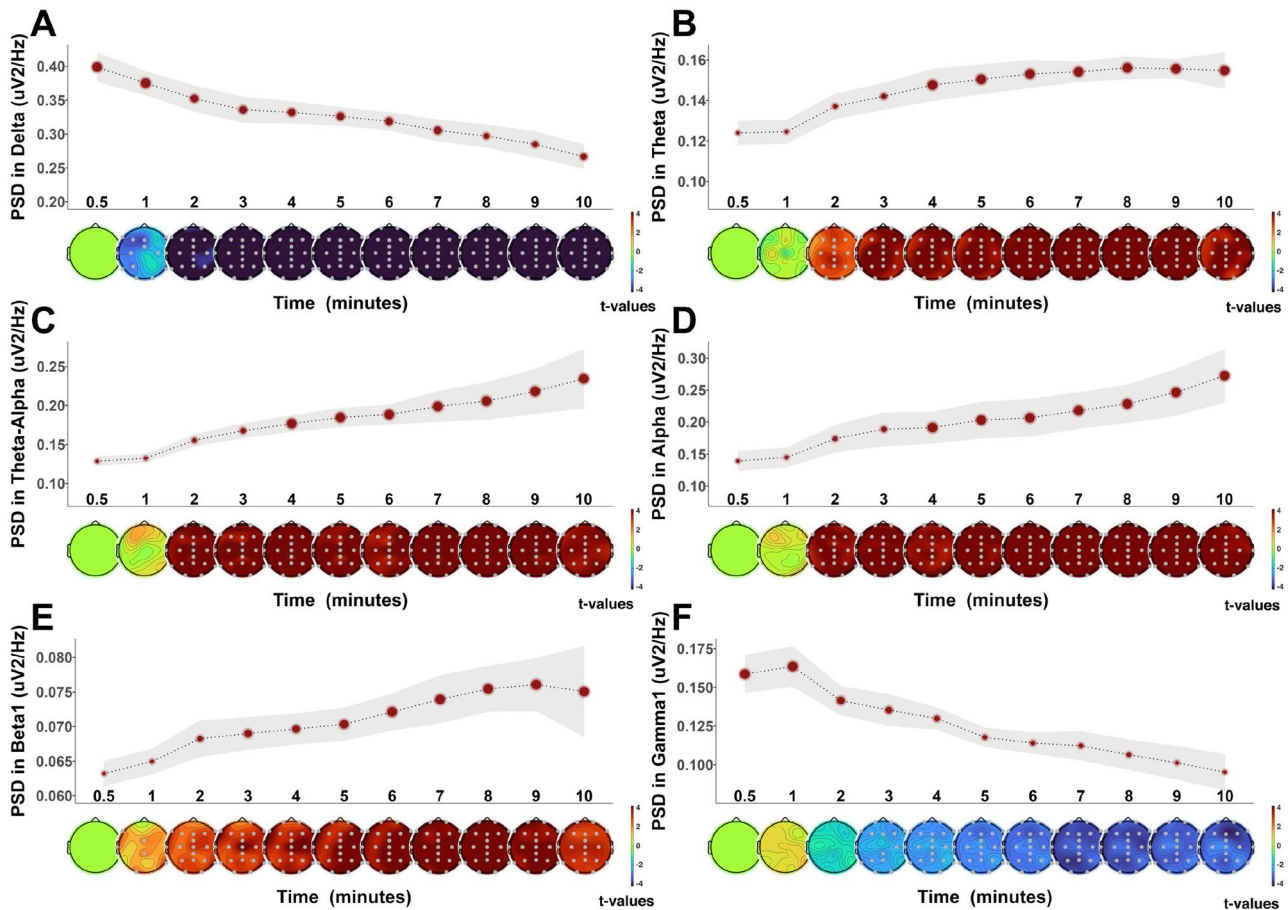


Fig. 2 Temporal dynamics in novice meditators: Time-to-onset and temporal dynamics during breath-watching meditation in novice meditators (NOV) are illustrated. Power-spectral values in different frequency bands (A: Delta; B: Theta; C: Theta-alpha; D: Alpha; E: Beta1; and F: Gamma1) show time-to-onset of effects around a 3-min duration. The size of the dots corresponds to the magnitude

of the values. The grey bar indicates standard error. The lower head plots are the t-value topographic plots (compared to the first 30 s) that highlight electrodes with statistically significant differences (dots, $p < 0.05$) in power. *P*-values have been corrected for multiple comparisons using cluster-based statistics

Finally, to justify the use of cumulative power analyses, we have added complementary analyses using distinct 30-s bins at each time point (Figs. S8–S14). These analyses yielded patterns consistent with the cumulative results: delta and gamma1 power decreased over time; theta, theta-alpha, and beta1 power increased over time; and no changes were observed in the beta2 band. Similar to the cumulative analyses, these changes emerged around the 2–3-min mark and peaked around 7–8 min.

Between-Group Comparisons

Our results showed significantly lower power in the delta band in advanced meditators at 0.5 min compared to novice

meditators ($p < 0.05$) and at 0.5- and 3 min compared to controls ($p < 0.05$) (Fig. 4A). Additionally, advanced meditators showed significantly higher theta power than both groups at the 0.5-, 3-, 6-, and 9-min durations ($p < 0.05$) (Fig. 4B). Similarly, advanced meditators showed significantly higher power in the theta-alpha band at 0.5-, 3-, and 6-min durations compared to controls ($p < 0.05$), and at 0.5 min compared to novice meditators ($p < 0.05$) (Fig. 4C). In the gamma frequency band, advanced meditators showed lower power than controls at 9 min ($p < 0.05$), with no significant differences observed at other time points (Fig. 4D). There were no statistically significant differences in other bands or comparisons between novices and controls (Figs. S15–S21 present box plots for all three groups at each time point).

Temporal dynamics during breath-watching meditation in advanced meditators

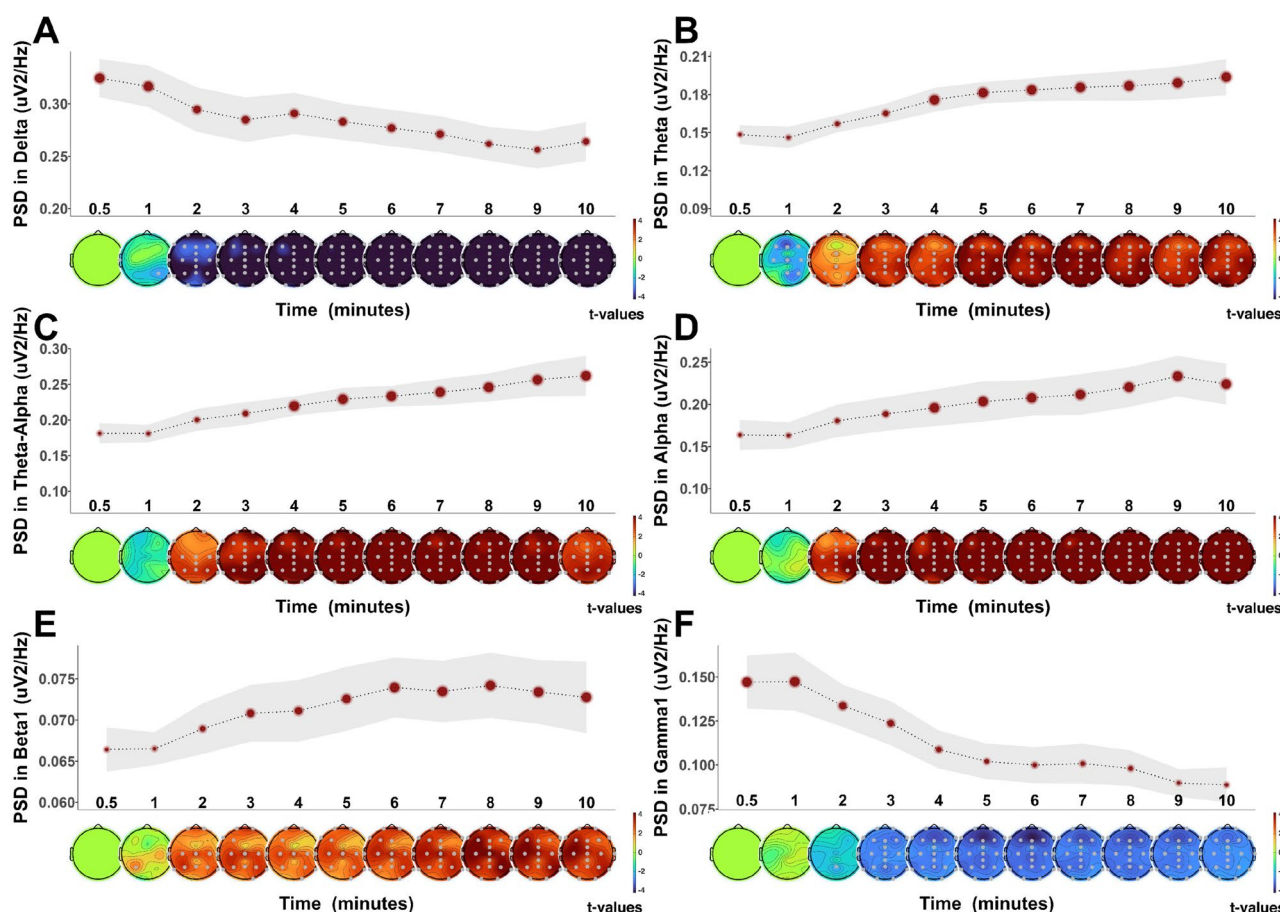


Fig. 3 Temporal dynamics in advanced meditators: Time-to-onset and temporal dynamics during breath-watching meditation in advanced meditators (ADV) are illustrated. Power-spectral values in different frequency bands (**A**: Delta; **B**: Theta; **C**: Theta-alpha; **D**: Alpha; **E**: Beta1; and **F**: Gamma1) show time-to-onset of effects around a 2-min duration. The size of the dots corresponds to the mag-

nitude of the values. The grey bar indicates standard error. The lower head plots are the *t*-value topographic plots (compared to the first 30 s) that highlight electrodes with statistically significant differences (dots, $p < 0.05$) in power. *P*-values have been corrected for multiple comparisons using cluster-based statistics

Thus, our results indicate that while advanced meditators do not exhibit marked differences in time-to-onset or peak EEG effects compared to other groups, the magnitude of these effects is greater. Specifically, advanced meditators showed stronger decreases in delta and gamma as well as stronger increases in alpha and theta compared to the other two groups. Finally, to understand how low-frequency (e.g., theta) and high-frequency (e.g., gamma1) oscillations covary over time, we calculated correlations between theta PSD and gamma1 PSD at each time point during breath-watching meditation in the three groups. Our results (Fig. 5, Fig. S22) showed a significant negative correlation ($p < 0.05$) between theta and gamma1 in all the three groups. Advanced meditators showed the strongest theta-gamma1 correlations, with coefficients in the high range (-0.65 to -0.74) that remained stable over time. Novice meditators showed the

weakest correlations, with coefficients in the low-to-medium range (-0.25 to -0.45) that appeared stable but occurred intermittently. Controls showed medium-to-high correlations (-0.48 to -0.75), which decreased till 5 min and then increasing linearly up to 10 min. These results represent a U-shaped pattern in theta-gamma1 correlations, with higher values in controls and advanced meditators and lower values in novices.

Overall, our results show: (1) decrease in delta and gamma over the course of 10 min meditation practice in all three groups starting around 2 min and peaking around 7–10 min; (2) increase in theta, theta-alpha, alpha, and beta1 power over the course of the 10 min meditation practice starting around 2 min and peaking around 7–10 min in all three groups; and (3) significantly different power values in

Temporal dynamics of brain power during breath-watching meditation

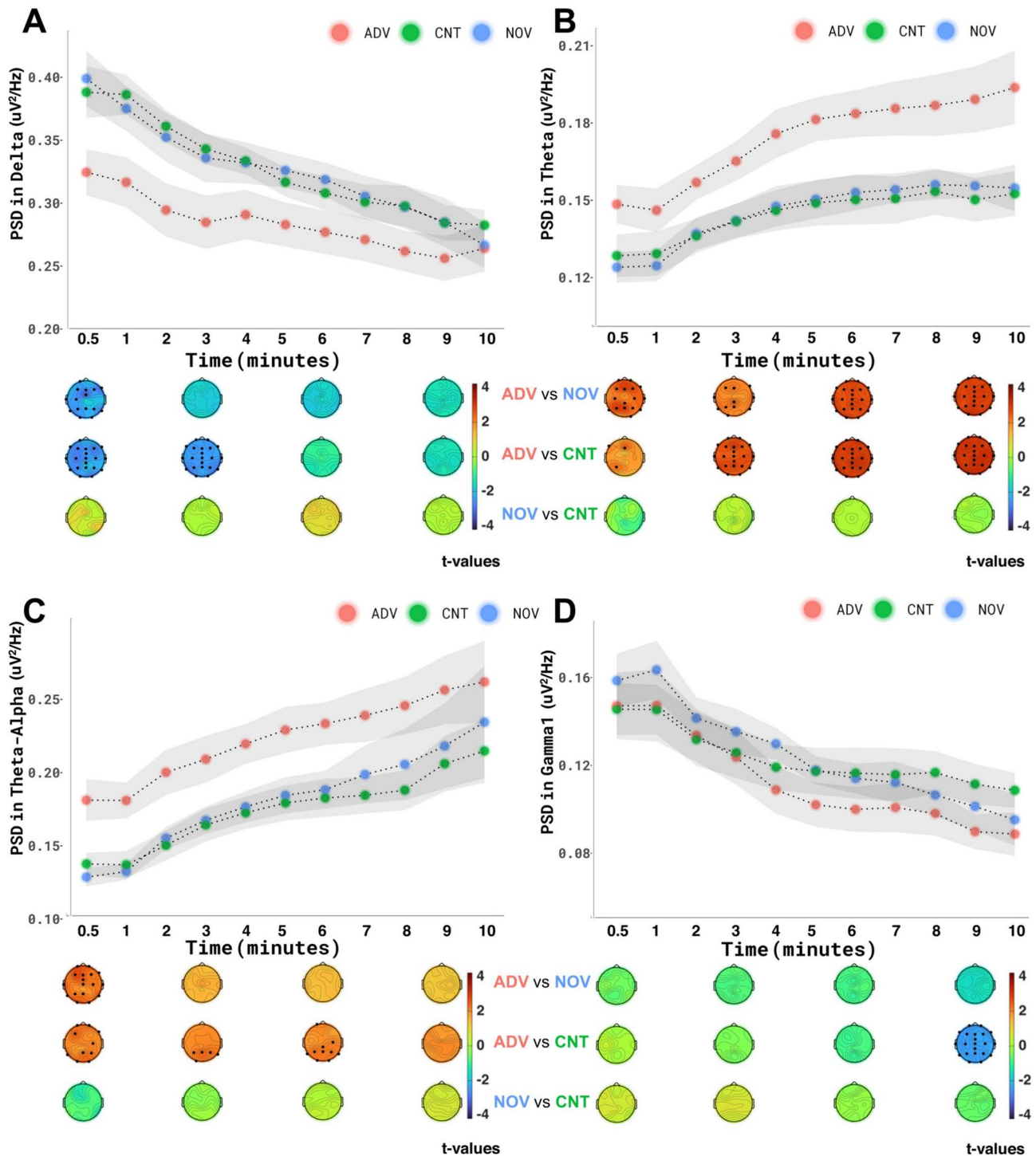
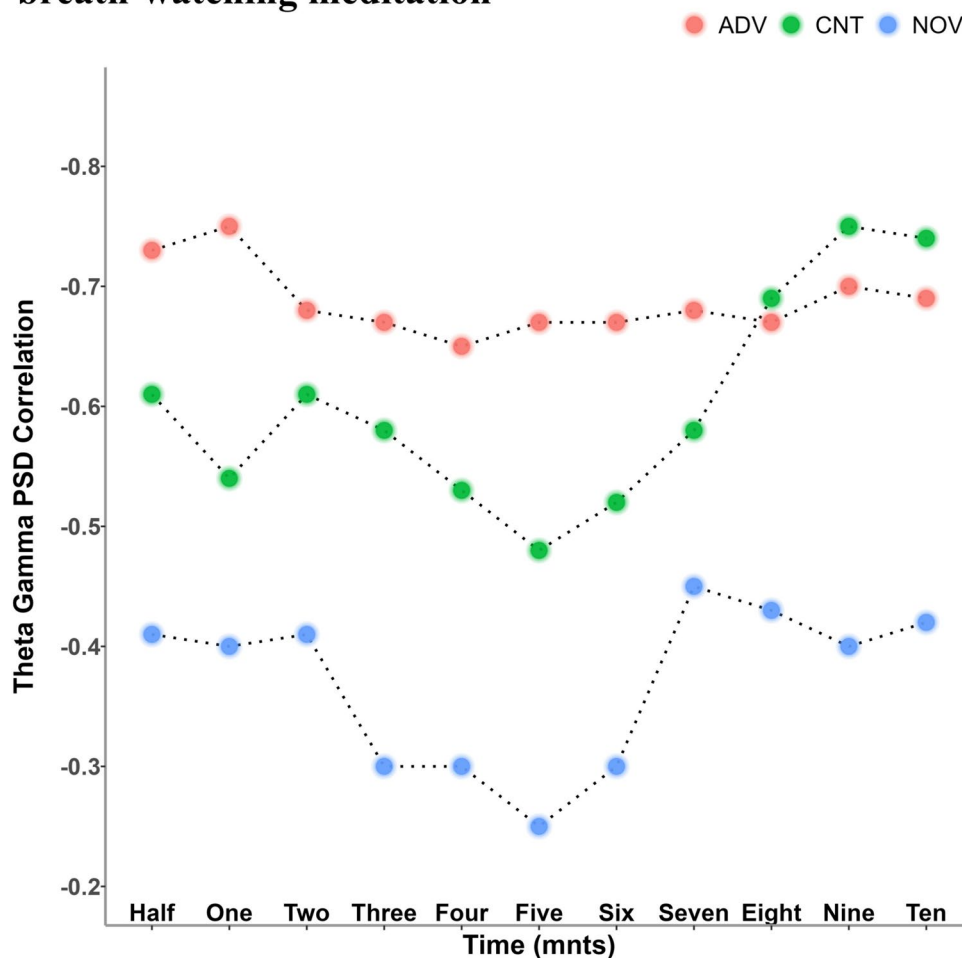


Fig. 4 Between group comparisons: Between-group comparisons of the temporal dynamics of neurophysiological effects across delta (A), theta (B), theta-alpha (C), and gamma1 (D) frequency bands during breath-watching meditation at 0.5, 3, 6, and 9 min. The grey bar shows the standard error of the mean. The lower head plots are the

t-value topographic plots that highlight electrodes with statistically significant differences (dots, $p < 0.05$) in power. P -values have been corrected for multiple comparisons using cluster-based statistics. Legend: ADV = Advanced meditators, NOV = Novice meditators, CNT = Meditation-naïve controls, PSD = Power-spectral density

Fig. 5 Temporal dynamics of Theta-gamma PSD correlation: The plot shows theta-gamma1 PSD correlation coefficients on the y-axis and time into breath-watching on the x-axis. Each dot represents the correlation coefficient obtained from the correlation between theta and gamma1 PSD values at each time point in advanced meditators (ADV), novice meditators (NOV), and controls (CNT)

Temporal dynamics of Theta-gamma correlation during breath-watching meditation



both delta/gamma (lower) and theta/theta-alpha (higher) in the advanced meditators compared to the other two groups.

Discussion

We studied the temporal dynamics (time-to-onset and peak effects) of EEG power spectra associated with breath-watching practice in advanced meditators, novice meditators, and controls. All groups showed significant brain changes, with some unique between-group differences, to occur around 2 min after starting meditation. Peak effects were observed between 7 and 10 min. These changes were expertise dependent: advanced meditators showed significantly greater power in the theta and theta-alpha bands, and significantly lower power in the delta and gamma1 bands compared to the other groups.

Across all groups, we found an increase in theta, theta-alpha, alpha, and beta1 power, and a decrease in delta and gamma1 power as breath-watching practice progressed. No

significant changes were found in the beta2 band. In controls, these effects consistently emerged at 2 min across all bands. In contrast, meditators displayed a more nuanced temporal pattern, with some effects appearing earlier — such as delta in novice meditators and theta in advanced meditators at 1 min — and others emerging later, such as gamma1 at 3 min in both meditator groups.

Enhanced theta, theta-alpha and alpha power are indicators of internalised attention and mental relaxation, widely reported across various meditation practices (Aftanas & Golocheikine, 2001; Brandmeyer & Delorme, 2018; Cahn & Polich, 2006; Fell et al., 2010; Kakumanu et al., 2018; Lomas et al., 2015). Greater beta1 power may reflect alert and focused attentional engagement (Britton et al., 2014b; Dunn et al., 1999; Lomas et al., 2015). A decrease in delta power may suggest increased alertness and reduced mind-wandering (Braboszcz & Delorme, 2011; Cahn et al., 2010; Caldwell et al., 2003; Cummins et al., 2024; Makeig & Inlow, 1993). Meanwhile, a decrease in gamma1 power is more difficult to interpret due to mixed findings in the

literature (Lee et al., 2018; Lomas et al., 2015) but may similarly reflect reduced mind-wandering (Cummins et al., 2024). While breath-watching meditation modulated activity in the beta1 and gamma1 bands, the lack of significant change in the beta2 band might be related to the nature of the practice, which promotes relaxed alertness (reflected in alpha and beta1 bands) without engaging the heightened alertness or the higher-order cognitive and sensorimotor processes typically associated with beta2 activity (Kropotov, 2016; Roger & Whittington, 2010). Overall, the observed pattern of EEG power changes may indicate that, as breath-watching progressed, participants became more stably focused on their breath with reduced mind-wandering (indicated by increases in theta and alpha, and a decrease in gamma1), and experienced a state of relaxed alertness (reflected by increases in theta, alpha, and beta1, and a decrease in delta), without signs of heightened alertness (as shown by no change in beta2 and a decrease in gamma1).

The unique temporal dynamics observed in the theta band among advanced meditators are particularly interesting. A decrease in theta power at 1 min followed by an increase at 2 min may reflect a dynamic topographic reorganization (Cooper et al., 2022) occurring in the early stages of meditation. Previous research has shown that advanced meditators can enter into a meditative state within a minute (Nair et al., 2017). We speculate that the initial decrease in theta could be driven by lower-frequency theta activity, which is more closely related to delta, while the subsequent increase may be driven by higher-frequency theta activity, which overlaps more with alpha. This pattern could reflect a transition toward a more alert and relaxed meditative state.

In addition to observing changes around the 2- to 3-min mark, we also found that peak neurophysiological effects occurred between 7 and 10 min, suggesting that it takes time to enter a deeper meditative state. Such insights cannot be obtained from static average-based analyses.

Given this understanding of temporal dynamics, how does meditation expertise relate to these changes? To explore this, we examined differences between practitioner groups. Advanced meditators showed significantly greater theta power at all time points, with higher theta power observed as early as 30 s. This indicates a trait feature of meditation (Cahn & Polich, 2006; Lomas et al., 2015; Malipeddi et al., 2024b). Meditation has been shown to have both state and trait effects (Cahn & Polich, 2006; Goleman & Davidson, 2018; Lomas et al., 2015). State effects of meditation are the immediate changes that occur during and immediately after a meditation session. These effects are often temporary and can vary from session to session (Cahn & Polich, 2006; Fell et al., 2010; Lee et al., 2018; Lomas et al., 2015). Whereas trait effects of meditation refer to the long-term changes that result from regular and consistent meditation practice. These effects are more

enduring and can become integrated into an individual's overall personality and way of being (Cahn & Polich, 2006; Goleman & Davidson, 2018; Lomas et al., 2015). However, the specific state and trait effects of meditation can vary among individuals, and the extent of these effects can depend on factors such as the duration and frequency of meditation practice, the techniques used, the intensity of practice, and other contextual factors such as personality, motivation, intentions, culture and teacher (Timmermann et al., 2022). Further, trait and state features interact, making it difficult to distinguish these – especially among the advanced meditators, where meditateness becomes a natural quality (Goleman & Davidson, 2018; Lutz et al., 2015; Tang et al., 2015). This could explain higher theta at 30 s in this group. Theta power increased linearly until 6 min before plateauing, whereas in the other two groups, plateauing occurred between 3 and 6 min. This could mean that advanced meditators achieved deeper states of meditation as time progressed. Theta power is a marker of expertise in meditation, with several studies reporting greater theta power in advanced meditators, compared to novice meditators, during meditation (Cahn & Polich, 2006; Fell et al., 2010; Lee et al., 2018; Lomas et al., 2015).

Higher power is linked to greater synchronization of neural oscillations (Buzsaki, 2006; Fell et al., 2010; Lutz et al., 2008). Given that breath-watching is a focused-attention practice, and advanced meditators are more trained in their attentional skills, attentional networks may be more synchronized, leading to greater theta power. In fact, in the previous research based on the same data, we showed that advanced meditators reported fewer hindrances (such as mind-wandering), higher levels of concentration (such as more focused attention), and an overall greater depth of meditation compared to the other groups (Malipeddi et al., 2024b). Overall, higher theta power in advanced meditators could indicate greater attention, meta-awareness, and pleasant states like bliss (Aftanas & Golocheikine, 2001; Brandmeyer & Delorme, 2021; Lomas et al., 2015; Tang et al., 2016).

We observed lower delta power in advanced meditators compared to the other groups during the first three min. A study on a focused breath-counting task observed an increase in delta power whenever subjects reported mind-wandering, which was interpreted as being linked to decreased alertness (Braboszcz & Delorme, 2011). Further, increased delta power has been linked to lower alertness in various other studies (Braboszcz & Delorme, 2011; Caldwell et al., 2003; De Gennaro et al., 2001; Makeig & Inlow, 1993). Our findings, thus, could potentially indicate significantly lower mind-wandering and more alertness in the advanced meditators. As time progressed, these effects disappeared, with all three groups reaching similar delta power levels, which could reflect similar alertness levels. Future studies should

incorporate experiential sampling and temporal dynamics together to better uncover these effects.

Lower gamma1 power observed in advanced meditators compared to controls at 9 min may indicate that, as time progresses, meditation deepens, leading to less mind-wandering and greater focus (Cummins et al., 2024). Interestingly, both advanced meditators and controls began with similar gamma1 power levels, but advanced meditators showed a greater decrease over time. No differences were found between advanced and novice meditators or between novice meditators and controls. This suggests that changes in the gamma1 band may require extensive meditation practice, aligning with previous studies in which gamma changes were mostly observed in advanced meditators (Braboszcz et al., 2017; Lomas et al., 2015; Lutz et al., 2004). However, this should be interpreted with caution. Notably, studies (Braboszcz et al., 2017; Lomas et al., 2015; Lutz et al., 2004) have reported increased gamma power in advanced meditators, contrary to our findings. This could be due to the gamma frequency studied. While prior studies have examined higher gamma, e.g., 60–110 Hz used by (Braboszcz et al., 2017), we studied lower gamma, 30–40 Hz. Additionally, the differences in gamma findings between ours and the literature may point to the use of the baseline and how it influences findings and interpretations.

While existing research has mostly used the eyes-closed resting state as their baseline (Cahn & Polich, 2006; Lee et al., 2018; Lomas et al., 2015), we employed the first 30 s of meditation practice as our baseline. We think this approach is justified, as our objective was to examine temporal effects, which required maintaining the same state as the baseline for consistency and relevance in the comparisons. Furthermore, the observation that initial neurophysiological changes occur around the 2–3-min mark in all groups supports the appropriateness of using the first 30 s as the baseline, as this choice did not result in missing any early practice-related changes, even amongst the advanced meditators.

A case in point that helps bring the importance of the baseline is in the contrast between our earlier findings (Malipeddi et al., 2024b) and the findings of this study. Using the eyes-closed resting state as the baseline, we previously showed that during breath-watching practice (see Fig. 4, Malipeddi et al., 2024b): delta and theta power increases in all groups; alpha power remains unchanged in advanced and novice meditators but decreases in controls; beta power increases in advanced and novice meditators but remains unchanged in controls; and gamma1 power increases marginally in advanced and novice meditators but remains unchanged in controls. However, using the first 30 s of the meditation state as the baseline, we found decreases in delta and gamma1 power over time and increases in theta, theta-alpha and beta1 power over time. This shows that using the appropriate baseline is important to interpret the findings

properly. Based on our findings, we suggest that future research should consider a temporal-dynamic approach to studying meditation, with the initial portion of the meditation state as the baseline, to better understand how neurophysiological changes evolve during meditation, particularly in high frequency gamma band.

Finally, PSD correlations between theta and gamma1 bands revealed interesting differences between the groups. Stronger negative correlation in these bands, that is stable over time, in advanced meditators might indicate a stable mental state characterized by sustained attention (elevated theta) and reduced sensory interference (suppressed gamma1). In contrast, the U-shaped pattern seen in novices and controls may indicate a mental state that fluctuates between focus and distraction. These interpretations remain speculative in the absence of direct reports of phenomenological state changes during meditation.

Practical Implications

Participation in long meditation retreats has been shown to significantly enhance well-being (Sadhasivam et al., 2020, 2021). However, this may not be practical or feasible for large segments of the population. Digital interventions that provide accessible meditation training through apps or online programs could help bridge this gap, making meditation more accessible to a wider population. In this context, our findings that just 3–7 min of meditation can induce significant neurophysiological changes, even in first-time meditators like our controls, are encouraging. Such brief meditation sessions could maximize benefits while minimizing common challenges faced by beginners, such as cognitive fatigue, boredom, and drowsiness. These findings support the integration of brief meditation practices into daily routines for mental health and cognitive benefits.

Mental health issues such as stress, anxiety, and depression are rising at an alarming rate worldwide (Jayasankar et al., 2022; Sagar et al., 2020; Santomauro et al., 2021; World Health Organization, n.d.). Research indicates that one in two individuals will develop a mental health disorder by the age of 75 (McGrath et al., 2023). In response to this growing crisis, the World Health Organization's Comprehensive Mental Health Action Plan 2013–2030 emphasizes the importance of investing in prevention of mental health conditions and promoting mental well-being (World Health Organization, 2021). Given growing scientific evidence supporting the role of meditation in promoting mental well-being (Creswell, 2017; Eberth & Sedlmeier, 2012; Galante et al., 2021; Hariri et al., 2022; Sedlmeier et al., 2012; Upadhyay, Joshi, et al., 2022), and the recent declaration of December 21st as World Meditation Day by the United Nations General Assembly (United Nations, n.d.), which underscores the relevance and importance of

meditation, our findings advocate for the global implementation and widespread adoption of simple meditative practices to nurture a healthy mind.

Limitations and Future Research

An important limitation of our study is the categorization of meditators into advanced and novice groups based solely on their reported practices. This does not consider other important contextual factors (Timmermann et al., 2022) such as personality and motivation that influence outcomes of meditation. Future studies should take these factors into account. Another important limitation of this study is the lack of phenomenological data on temporal changes in meditative depth and quality, which could have complemented the neurophysiological findings and offered deeper insights into the temporal dynamics of meditation. Additionally, the absence of pre- and post-meditation mental well-being measures limited us from correlating frequency band changes with subjective reports of mental well-being. Another limitation of this study is the lack of data on autonomic function, such as heart rate variability, which is a well-established marker of relaxation commonly observed during meditation. Additionally, this study captures only the short-term effects of a single meditation session; the absence of longitudinal follow-up prevents us from assessing whether these neurophysiological changes translate into sustained, long-term benefits with continued practice. Lastly, this study may be subject to selection bias, as individuals with positive attitudes toward meditation are more likely to volunteer for such research, potentially influencing the outcomes.

Our study addresses a critical gap in the literature by providing insights into when neurophysiological changes begin and peak during meditation. We observed that these changes start around 2–3 min and peak around 7–10 min across all groups during breath-watching meditation. These changes included increased alpha, theta, and beta1 power and decreased delta and gamma1 power, with no changes in beta2. Notably, while meditation expertise did not alter the timing of these changes, it significantly influenced their magnitude, with advanced meditators showing greater theta and theta-alpha power, and lower delta and gamma power. Overall, we suggest that the brain's response to meditation can be rapid and varies with the practitioner's experience, potentially influencing cognitive and emotional processing in significant ways. Our findings offer novel insights into the temporal dynamics of EEG changes during meditation. The study highlights the potential of personalized meditation programs that are tailored to individuals' experience levels and consider factors such as age, gender, and cultural background, thereby optimizing mental health outcomes.

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Data Availability Data and code supporting the findings of this study will be made available upon request. For access, please contact [Dr. Saketh Malipeddi](mailto:sakethreddy21@gmail.com).

Declarations

Competing interests The authors declare no competing interests.

Ethics Approval The study was reviewed and approved by the Institute Ethics Committee (IEC) of the National Institute of Mental Health and Neurosciences (NIMHANS) (Ethics Approval: NIMH/DO/ETHICS SUB-COMMITTEE MEETING/2018), in accordance with the ethical guidelines for biomedical research on human participants issued by the Indian Council of Medical Research (ICMR).

Informed Consent All of the participants provided written informed consent prior to participation.

Use of Artificial Intelligence ChatGPT (version 5.2) was used solely for language editing and grammatical refinement. No AI tools were used in the generation of the scientific content of this manuscript.

References

- Aftanas, L. I., & Golocheikine, S. A. (2001). Human anterior and frontal midline theta and lower alpha reflect emotionally positive state and internalized attention: High-resolution EEG investigation of meditation. *Neuroscience Letters*, 310(1), 57–60. [https://doi.org/10.1016/s0304-3940\(01\)02094-8](https://doi.org/10.1016/s0304-3940(01)02094-8)
- Braboszcz, C., Cahn, B. R., Balakrishnan, B., Maturi, R. K., Grandchamp, R., & Delorme, A. (2013). Plasticity of visual attention in Isha yoga meditation practitioners before and after a 3-month retreat. *Frontiers in Psychology*, 4, 914. <https://doi.org/10.3389/fpsyg.2013.00914>
- Braboszcz, C., Cahn, B. R., Levy, J., Fernandez, M., & Delorme, A. (2017). Increased gamma brainwave amplitude compared to control in three different meditation traditions. *PLoS ONE*, 12(1), e0170647. <https://doi.org/10.1371/journal.pone.0170647>
- Braboszcz, C., & Delorme, A. (2011). Lost in thoughts: Neural markers of low alertness during mind wandering. *NeuroImage*, 54(4), 3040–3047. <https://doi.org/10.1016/j.neuroimage.2010.10.008>
- Brandmeyer, T., & Delorme, A. (2018). Reduced mind wandering in experienced meditators and associated EEG correlates.

- Experimental Brain Research*, 236, 2519–2528. <https://doi.org/10.1007/s00221-016-4811-5>
- Brandmeyer, T., & Delorme, A. (2021). Meditation and the wandering mind: A theoretical framework of underlying neurocognitive mechanisms. *Perspectives on Psychological Science*, 16(1), 39–66. <https://doi.org/10.1177/1745691620917340>
- Britton, W. B., Lindahl, J. R., Cahn, B. R., Davis, J. H., & Goldman, R. E. (2014). Awakening is not a metaphor: The effects of Buddhist meditation practices on basic wakefulness. *Annals of the New York Academy of Sciences*, 1307(1), 64–81. <https://doi.org/10.1111/nyas.12279>
- Britton, W. B., Lindahl, J. R., Cahn, B. R., Davis, J. H., & Goldman, R. E. (2014). Awakening is not a metaphor: The effects of Buddhist meditation practices on basic wakefulness: Meditation and wakefulness. *Annals of the New York Academy of Sciences*, 1307(1), 64–81. <https://doi.org/10.1111/nyas.12279>
- Buzsaki, G. (2006). *Rhythms of the brain*. Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780195301069.001.0001>
- Cahn, B. R., Delorme, A., & Polich, J. (2010). Occipital gamma activation during Vipassana meditation. *Cognitive Processing*, 11(1), 39–56. <https://doi.org/10.1007/s10339-009-0352-1>
- Cahn, B. R., Goodman, M. S., Peterson, C. T., Maturi, R., & Mills, P. J. (2017). Yoga, meditation and mind-body health: Increased BDNF, cortisol awakening response, and altered inflammatory marker expression after a 3-month yoga and meditation retreat. *Frontiers in Human Neuroscience*, 11, 315. <https://doi.org/10.3389/fnhum.2017.00315>
- Cahn, B. R., & Polich, J. (2006). Meditation states and traits: EEG, ERP, and neuroimaging studies. *Psychological Bulletin*, 132, 180–211. <https://doi.org/10.1037/0033-2909.132.2.180>
- Caldwell, J. A., Prazinko, B., & Caldwell, J. L. (2003). Body posture affects electroencephalographic activity and psychomotor vigilance task performance in sleep-deprived subjects. *Clinical Neurophysiology*, 114(1), 23–31. [https://doi.org/10.1016/S1388-2457\(02\)00283-3](https://doi.org/10.1016/S1388-2457(02)00283-3)
- Chandran, V., Bermúdez, M.-L., Koka, M., Chandran, B., Pawale, D., Vishnubhotla, R., Alankar, S., Maturi, R., Subramaniam, B., & Sadhasivam, S. (2021). Large-scale genomic study reveals robust activation of the immune system following advanced Inner Engineering meditation retreat. *Proceedings of the National Academy of Sciences of the United States of America*, 118(51), e2110455118. <https://doi.org/10.1073/pnas.2110455118>
- Chang, T. F. H. (2021). An emerging positive intervention—Inner Engineering Online (IEO). *Journal of Occupational and Environmental Medicine*, 63(9), e657. <https://doi.org/10.1097/JOM.0000000000002270>
- Chang, T. F. H., Ley, B. L., Ramburn, T. T., Srinivasan, S., Hariri, S., Purandare, P., & Subramaniam, B. (2022). Online Isha Upa Yoga for student mental health and well-being during COVID-19: A randomized control trial. *Applied Psychology: Health and Well-Being*, 14(4), 1408–1428. <https://doi.org/10.1111/aphw.12341>
- Cooper, A. C., Ventura, B., & Northoff, G. (2022). Beyond the veil of duality—Topographic reorganization model of meditation. *Neuroscience of Consciousness*, 2022(1), niac013. <https://doi.org/10.1093/nc/niac013>
- Creswell, J. D. (2017). Mindfulness interventions. *Annual Review of Psychology*, 68(1), 491–516. <https://doi.org/10.1146/annurev-psych-042716-051139>
- Cummins, D. D., Schulman, Z., Maher, C., Tortolero, L., Saad, A., Nunez Martinez, L., Davidson, R. J., Marcuse, L. V., Saez, I., & Panov, F. (2024). Influence of mindfulness meditation on intracranial EEG parameters in epileptic and non-epileptic brain areas. *Epilepsy & Behavior*, 161, 110150. <https://doi.org/10.1016/j.yebeh.2024.110150>
- Dahl, C. J., Lutz, A., & Davidson, R. J. (2015). Reconstructing and deconstructing the self: Cognitive mechanisms in meditation practice. *Trends in Cognitive Sciences*, 19(9), 515–523. <https://doi.org/10.1016/j.tics.2015.07.001>
- Dahl, C. J., Wilson-Mendenhall, C. D., & Davidson, R. J. (2020). The plasticity of well-being: A training-based framework for the cultivation of human flourishing. *Proceedings of the National Academy of Sciences of the United States of America*, 117(51), 32197–32206. <https://doi.org/10.1073/pnas.2014859117>
- Davies, J. N., Faschinger, A., Galante, J., & Van Dam, N. T. (2024). Prevalence and 20-year trends in meditation, yoga, guided imagery and progressive relaxation use among US adults from 2002 to 2022. *Scientific Reports*, 14(1), 14987. <https://doi.org/10.1038/s41598-024-64562-y>
- De Gennaro, L., Ferrara, M., & Bertini, M. (2001). The boundary between wakefulness and sleep: Quantitative electroencephalographic changes during the sleep onset period. *Neuroscience*, 107(1), 1–11. [https://doi.org/10.1016/S0306-4522\(01\)00309-8](https://doi.org/10.1016/S0306-4522(01)00309-8)
- Delorme, A., & Makeig, S. (2004). EEGLAB: An open source toolbox for analysis of single-trial EEG dynamics including independent component analysis. *Journal of Neuroscience Methods*, 134(1), 9–21. <https://doi.org/10.1016/j.jneumeth.2003.10.009>
- Dunn, B. R., Hartigan, J. A., & Mikulas, W. L. (1999). Concentration and mindfulness meditations: Unique forms of consciousness? *Applied Psychophysiology and Biofeedback*, 24(3), 147–165. <https://doi.org/10.1023/a:1023498629385>
- Eberth, J., & Sedlmeier, P. (2012). The effects of mindfulness meditation: A meta-analysis. *Mindfulness*, 3(3), 174–189. <https://doi.org/10.1007/s12671-012-0101-x>
- Fell, J., Axmacher, N., & Haupt, S. (2010). From alpha to gamma: Electrophysiological correlates of meditation-related states of consciousness. *Medical Hypotheses*, 75(2), 218–224. <https://doi.org/10.1016/j.mehy.2010.02.025>
- Fox, K. C. R., Dixon, M. L., Nijeboer, S., Girn, M., Floman, J. L., Lifshitz, M., Ellamil, M., Sedlmeier, P., & Christoff, K. (2016). Functional neuroanatomy of meditation: A review and meta-analysis of 78 functional neuroimaging investigations. *Neuroscience and Biobehavioral Reviews*, 65, 208–228. <https://doi.org/10.1016/j.neubiorev.2016.03.021>
- Fox, K. C. R., Nijeboer, S., Dixon, M. L., Floman, J. L., Ellamil, M., Rumak, S. P., Sedlmeier, P., & Christoff, K. (2014). Is meditation associated with altered brain structure? A systematic review and meta-analysis of morphometric neuroimaging in meditation practitioners. *Neuroscience and Biobehavioral Reviews*, 43, 48–73. <https://doi.org/10.1016/j.neubiorev.2014.03.016>
- Galante, J., Friedrich, C., Dawson, A. F., Modrego-Alarcón, M., Gebbing, P., Delgado-Suárez, I., Gupta, R., Dean, L., Dalgleish, T., White, I. R., & Jones, P. B. (2021). Mindfulness-based programmes for mental health promotion in adults in nonclinical settings: A systematic review and meta-analysis of randomised controlled trials. *PLoS Medicine*, 18(1), e1003481. <https://doi.org/10.1371/journal.pmed.1003481>
- Ganesan, S., Beyer, E., Moffat, B., Van Dam, N. T., Lorenzetti, V., & Zalesky, A. (2022). Focused attention meditation in healthy adults: A systematic review and meta-analysis of cross-sectional functional MRI studies. *Neuroscience and Biobehavioral Reviews*, 141, 104846. <https://doi.org/10.1016/j.neubiorev.2022.104846>
- Goleman, D., & Davidson, R. J. (2018). *Altered traits: Science reveals how meditation changes your mind, brain, and body*. Penguin.
- Goyal, M., Singh, S., Sibinga, E. M. S., Gould, N. F., Rowland-Seymour, A., Sharma, R., Berger, Z., Sleicher, D., Maron, D. D., Shihab, H. M., Ranasinghe, P. D., Linn, S., Saha, S., Bass, E. B., & Haythornthwaite, J. A. (2014). Meditation programs for psychological stress and well-being: A systematic review and

- meta-analysis. *JAMA Internal Medicine*, 174(3), 357–368. <https://doi.org/10.1001/jamainternmed.2013.13018>
- Hariri, S., Vishnubhotla, R. V., Reed, P. U., Rayapuraj, A., Orui, H., Balachundhar, P., Sadhasivam, S., & Subramaniam, B. (2022). Online guided meditation training (Isha Kriya) improves self-reported symptoms of anxiety and depression within 2 weeks of practice—An observational study. *Frontiers in Psychiatry* 13, 944973. <https://doi.org/10.3389/fpsy.2022.944973>
- Hölzel, B. K., Lazar, S. W., Gard, T., Schuman-Olivier, Z., Vago, D. R., & Ott, U. (2011). How does mindfulness meditation work? Proposing mechanisms of action from a conceptual and neural perspective. *Perspectives on Psychological Science*, 6(6), 537–559. <https://doi.org/10.1177/1745691611419671>
- Jayasankar, P., Manjunatha, N., Rao, G. N., Gururaj, G., Varghese, M., Benegal, V., NMHS India National Collaborator Group. (2022). Epidemiology of common mental disorders: Results from “National Mental Health Survey” of India, 2016. *Indian Journal of Psychiatry*, 64(1), 13–19. https://doi.org/10.4103/indianjpsychiatry.indianjpsychiatry_865_21
- Jha, A. P., Krompinger, J., & Baime, M. J. (2007). Mindfulness training modifies subsystems of attention. *Cognitive, Affective & Behavioral Neuroscience*, 7(2), 109–119. <https://doi.org/10.3758/CABN.7.2.109>
- Kakumanu, R. J., Nair, A. K., Venugopal, R., Sasidharan, A., Ghosh, P. K., John, J. P., Mehrotra, S., Panth, R., & Kutty, B. M. (2018). Dissociating meditation proficiency and experience dependent EEG changes during traditional Vipassana meditation practice. *Biological Psychology*, 135, 65–75. <https://doi.org/10.1016/j.biopsycho.2018.03.004>
- Krol, L. R. (2020). *EEG electrode positions in the 10–10 system using modified combinatorial nomenclature*. [Graphic]. Own work. https://commons.wikimedia.org/wiki/File:EEG_10-10_system_with_additional_information.svg
- Kropotov, J. D. (2016). Chapter 2.3—Beta and Gamma Rhythms. In J. D. Kropotov (Ed.), *Functional Neuromarkers for Psychiatry* (pp. 107–119). Academic Press. <https://doi.org/10.1016/B978-0-12-410513-3.00009-7>
- Lee, D. J., Kulubya, E., Goldin, P., Goodarzi, A., & Girgis, F. (2018). Review of the neural oscillations underlying meditation. *Frontiers in Neuroscience*, 12, 178. <https://doi.org/10.3389/fnins.2018.00178>
- Lomas, T., Ivtzan, I., & Fu, C. H. Y. (2015). A systematic review of the neurophysiology of mindfulness on EEG oscillations. *Neuroscience and Biobehavioral Reviews*, 57, 401–410. <https://doi.org/10.1016/j.neubiorev.2015.09.018>
- Lutz, A., Greischar, L. L., Rawlings, N. B., Ricard, M., & Davidson, R. J. (2004). Long-term meditators self-induce high-amplitude gamma synchrony during mental practice. *Proceedings of the National Academy of Sciences*, 101(46), 16369–16373. <https://doi.org/10.1073/pnas.0407401101>
- Lutz, A., Jha, A. P., Dunne, J. D., & Saron, C. D. (2015). Investigating the phenomenological matrix of mindfulness-related practices from a neurocognitive perspective. *American Psychologist*, 70(7), 632–658. <https://doi.org/10.1037/a0039585>
- Lutz, A., Slagter, H. A., Dunne, J. D., & Davidson, R. J. (2008). Attention regulation and monitoring in meditation. *Trends in Cognitive Sciences*, 12(4), 163–169. <https://doi.org/10.1016/j.tics.2008.01.005>
- Mair, P., & Wilcox, R. (2020). Robust statistical methods in R using the WRS2 package. *Behavior Research Methods*, 52(2), 464–488. <https://doi.org/10.3758/s13428-019-01246-w>
- Makeig, S., & Inlow, M. (1993). Lapse in alertness: Coherence of fluctuations in performance and EEG spectrum. *Electroencephalography and Clinical Neurophysiology*, 86(1), 23–35. [https://doi.org/10.1016/0013-4694\(93\)90064-3](https://doi.org/10.1016/0013-4694(93)90064-3)
- Malipeddi, S., Mehrotra, S., John, J. P., & Kutty, B. M. (2024). Practice and proficiency of Isha Yoga for better mental health outcomes: Insights from a COVID-19 survey. *Frontiers in Public Health*, 12, 1280859. <https://doi.org/10.3389/fpubh.2024.1280859>
- Malipeddi, S., Sasidharan, A., P.N., R., Mehrotra, S., John, J. P., & Kutty, B. M. (2024b). Relaxed alertness in novice and advanced meditators – A neurophysiological and psychological study of Isha Yoga practices. *Mindfulness*, 15(11), 2840–2859. <https://doi.org/10.1007/s12671-024-02466-8>
- McGrath, J. J., Al-Hamzawi, A., Alonso, J., Altwaijri, Y., Andrade, L. H., Bromet, E. J., Bruffaerts, R., de Almeida, J. M. C., Char-doul, S., Chiu, W. T., Degenhardt, L., Demler, O. V., Ferry, F., Gureje, O., Haro, J. M., Karam, E. G., Karam, G., Khaled, S. M., Kovess-Masfety, V., ... Zaslavsky, A. M. (2023). Age of onset and cumulative risk of mental disorders: A cross-national analysis of population surveys from 29 countries. *The Lancet Psychiatry*, 10(9), 668–681. [https://doi.org/10.1016/S2215-0366\(23\)00193-1](https://doi.org/10.1016/S2215-0366(23)00193-1)
- Muralikrishnan, K., Balakrishnan, B., Balasubramanian, K., & Vis-negarawla, F. (2012). Measurement of the effect of Isha Yoga on cardiac autonomic nervous system using short-term heart rate variability. *Journal of Ayurveda and Integrative Medicine*, 3(2), 91–96. <https://doi.org/10.4103/0975-9476.96528>
- Muthukumaraswamy, S. D. (2013). High-frequency brain activity and muscle artifacts in MEG/EEG: A review and recommendations. *Frontiers in Human Neuroscience*, 7, 138. <https://doi.org/10.3389/fnhum.2013.00138>
- Nair, A. K., Sasidharan, A., John, J. P., Mehrotra, S., & Kutty, B. M. (2017). Just a minute meditation: Rapid voluntary conscious state shifts in long term meditators. *Consciousness and Cognition*, 53, 176–184. <https://doi.org/10.1016/j.concog.2017.06.002>
- Nanamoli, B. (2010). *Mindfulness of breathing: Anapanasati*. Buddhist Publication Society.
- NCCIH. (n.d.). *Meditation and Mindfulness: What You Need To Know*. NCCIH. Retrieved August 5, 2024, from <https://www.nccih.nih.gov/health/meditation-and-mindfulness-what-you-need-to-know>
- Pernet, C. R., Chauveau, N., Gaspar, C., & Rousselet, G. A. (2011). LIMO EEG: A toolbox for hierarchical Linear Modeling of electroencephalographic data. *Computational Intelligence and Neuroscience*. <https://doi.org/10.1155/2011/831409>
- Pernet, C. R., Latinus, M., Nichols, T. E., & Rousselet, G. A. (2015). Cluster-based computational methods for mass univariate analyses of event-related brain potentials/fields: A simulation study. *Journal of Neuroscience Methods*, 250, 85–93. <https://doi.org/10.1016/j.jneumeth.2014.08.003>
- Peterson, C. T., Bauer, S. M., Chopra, D., Mills, P. J., & Maturi, R. K. (2017). Effects of Shambhavi Mahamudra Kriya, a multicomponent breath-based yogic practice (pranayama), on perceived stress and general well-being. *Journal of Evidence-Based Complementary & Alternative Medicine*, 22(4), 788–797. <https://doi.org/10.1177/2156587217730934>
- Rahul, V. (2023). *Rahulvenugopal/WhoIsYourNeighbour* [MATLAB]. <https://github.com/rahulvenugopal/WhoIsYourNeighbour> (Original work published 2023)
- Raman, M., Vishnubhotla, R., Ramay, H. R., Gonçalves, M. C. B., Shin, A. S., Pawale, D., Subramaniam, B., & Sadhasivam, S. (2023). Isha yoga practices, vegan diet, and participation in Samyama meditation retreat: Impact on the gut microbiome & metabolome - a non-randomized trial. *BMC Complementary Medicine and Therapies*, 23(1), 107. <https://doi.org/10.1186/s12906-023-03935-8>
- Roger, R. D., & Whittington, M. A. (2010). Beta-2 Oscillations. In M. Traub Roger & P. Whittington Miles (Eds.), *Cortical Oscillations in Health and Disease* (p. 0). Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780195342796.003.0011>
- Sadhasivam, S., Alankar, S., Maturi, R., Vishnubhotla, R. V., Mudigonda, M., Pawale, D., Narayanan, S., Hariri, S., Ram, C.,

- Chang, T., Renschler, J., Eckert, G., & Subramaniam, B. (2020). Inner Engineering practices and advanced 4-day Isha yoga retreat are associated with cannabinimimetic effects with increased endocannabinoids and short-term and sustained improvement in mental health: A prospective observational study of meditators. *Evidence-Based Complementary and Alternative Medicine*, 2020, 8438272. <https://doi.org/10.1155/2020/8438272>
- Sadhasivam, S., Alankar, S., Maturi, R., Williams, A., Vishnubhotla, R. V., Hariri, S., Mudigonda, M., Pawale, D., Dubbireddi, S., Packiasabapathy, S., Castelluccio, P., Ram, C., Renschler, J., Chang, T., & Subramaniam, B. (2021). Isha Yoga practices and participation in Samyama program are associated with reduced HbA1C and systemic inflammation, improved lipid profile, and short-term and sustained improvement in mental health: A prospective observational study of meditators. *Frontiers in Psychology*, 12, 659667. <https://doi.org/10.3389/fpsyg.2021.659667>
- Sadhguru, J. V. (2016). *Inner engineering*. Penguin Random House.
- Sagar, R., Dandona, R., Gururaj, G., Dhaliwal, R. S., Singh, A., Ferrari, A., Dua, T., Ganguli, A., Varghese, M., Chakma, J. K., Kumar, G. A., Shaji, K. S., Ambekar, A., Rangaswamy, T., Vijayakumar, L., Agarwal, V., Krishnakutty, R. P., Bhatia, R., Charlson, F., ... Dandona, L. (2020). The burden of mental disorders across the states of India: The Global Burden of Disease Study 1990–2017. *The Lancet Psychiatry*, 7(2), 148–161. [https://doi.org/10.1016/S2215-0366\(19\)30475-4](https://doi.org/10.1016/S2215-0366(19)30475-4)
- Santomauro, D. F., Mantilla Herrera, A. M., Shadid, J., Zheng, P., Ashbaugh, C., Pigott, D. M., Abbafati, C., Adolph, C., Amlag, J. O., Aravkin, A. Y., Bang-Jensen, B. L., Bertolacci, G. J., Bloom, S. S., Castellano, R., Castro, E., Chakrabarti, S., Chattopadhyay, J., Cogen, R. M., Collins, J. K., ... Ferrari, A. J. (2021). Global prevalence and burden of depressive and anxiety disorders in 204 countries and territories in 2020 due to the COVID-19 pandemic. *Lancet*, 398(10312), 1700–1712. [https://doi.org/10.1016/S0140-6736\(21\)02143-7](https://doi.org/10.1016/S0140-6736(21)02143-7)
- Sedlmeier, P., Eberth, J., Schwarz, M., Zimmermann, D., Haarig, F., Jaeger, S., & Kunze, S. (2012). The psychological effects of meditation: A meta-analysis. *Psychological Bulletin*, 138(6), 1139–1171. <https://doi.org/10.1037/a0028168>
- Selvaraj, N., Shivplara, N. B., Bhatia, M., Santhosh, J., Deepak, K. K., & Anand, S. (2008). Heart Rate Dynamics during Shambhavi Mahamudra- A Practice of Isha Yoga. *Journal of Complementary and Integrative Medicine*, 5(1). <https://doi.org/10.2202/1553-3840.1137>
- Smith, S. M., & Nichols, T. E. (2009). Threshold-free cluster enhancement: Addressing problems of smoothing, threshold dependence and localisation in cluster inference. *NeuroImage*, 44(1), 83–98. <https://doi.org/10.1016/j.neuroimage.2008.03.061>
- Sylapan, B. S., Nair, A. K., Jayanna, K., Mallipeddi, S., Sathyanarayana, S., & Kutty, B. M. (2020). Meditation, well-being and cognition in heartfulness meditators—A pilot study. *Consciousness and Cognition*, 86, 103032. <https://doi.org/10.1016/j.concog.2020.103032>
- Tang, Y.-Y., Hölzel, B. K., & Posner, M. I. (2015). The neuroscience of mindfulness meditation. *Nature Reviews Neuroscience*, 16(4), 4. <https://doi.org/10.1038/nrn3916>
- Tang, Y.-Y., Hölzel, B. K., & Posner, M. I. (2016). Traits and states in mindfulness meditation. *Nature Reviews Neuroscience*, 17(1), 1. <https://doi.org/10.1038/nrn.2015.7>
- Timmermann, C., Bauer, P. R., Gosseries, O., Vanhaudenhuyse, A., Vollenweider, F., Laureys, S., Singer, T., Antonova, E., & Lutz, A. (2022). A neurophenomenological approach to non-ordinary states of consciousness: Hypnosis, meditation, and psychedelics. *Trends in Cognitive Sciences*, 27(2), 139–159. <https://doi.org/10.1016/j.tics.2022.11.006>
- United Nations. (n.d.). *World Meditation Day*. United Nations; United Nations. Retrieved January 12, 2025, from <https://www.un.org/en/observances/meditation-day>
- Upadhyay, P., Joshi, A., Mishra, I., Kelly, L., Novack, L., Hariri, S., Kveraga, K., & Subramaniam, B. (2022). Short term effects of Inner Engineering completion online program on stress and well-being measures. *Frontiers in Psychology*, 13, 814224. <https://doi.org/10.3389/fpsyg.2022.814224>
- Upadhyay, P., Narayanan, S., Khera, T., Kelly, L., Mathur, P. A., Shanker, A., Novack, L., Pérez-Robles, R., Hoffman, K. A., Sadhasivam, S. K., & Subramaniam, B. (2022b). Perceived stress, resilience, and wellbeing in seasoned Isha Yoga practitioners compared to matched controls during the COVID-19 pandemic. *Frontiers in Public Health*, 10, 813664. <https://doi.org/10.3389/fpubh.2022.813664>
- Vishnubhotla, R. V., Radhakrishnan, R., Kveraga, K., Deardorff, R., Ram, C., Pawale, D., Wu, Y.-C., Renschler, J., Subramaniam, B., & Sadhasivam, S. (2021). Advanced meditation alters resting-state brain network connectivity correlating with improved mindfulness. *Frontiers in Psychology*, 12, 745344. <https://doi.org/10.3389/fpsyg.2021.745344>
- World Health Organization. (2021). *Comprehensive mental health action plan 2013–2030*. World Health Organization.
- World Health Organization. (n.d.). *Mental disorders*. <https://www.who.int/news-room/fact-sheets/detail/mental-disorders>

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